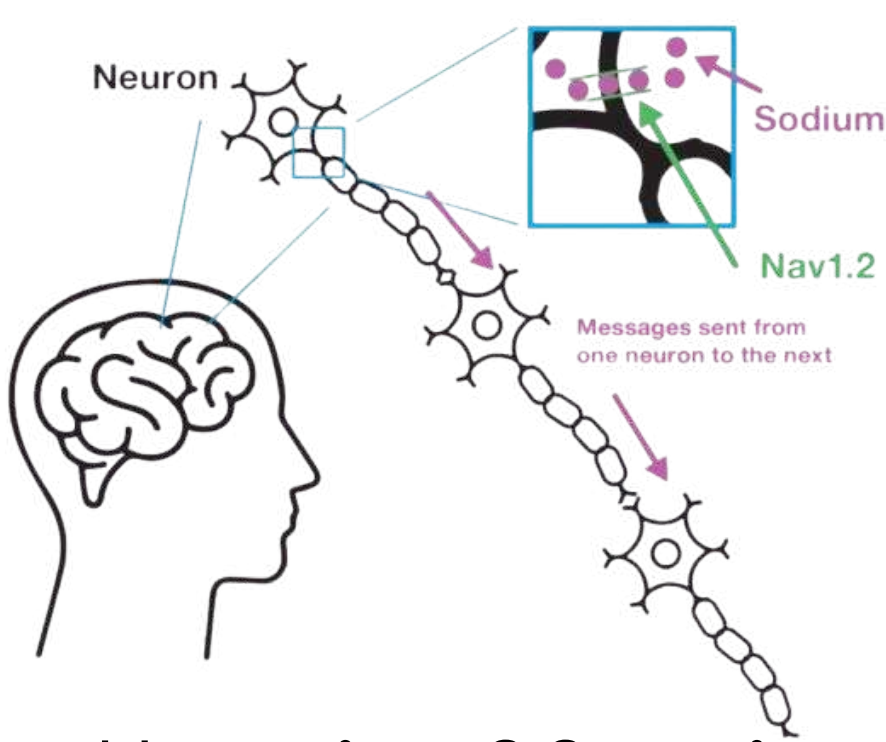


INVESTIGATING THE EFFECTS OF LINALLOOL ON NAV1.2 PROTEIN CHANNEL CALCIUM INFLUX OF SH-SY5Y TRANSFECTED CELLS

Category: CELLULAR & MOLECULAR BIOLOGY

BACKGROUND

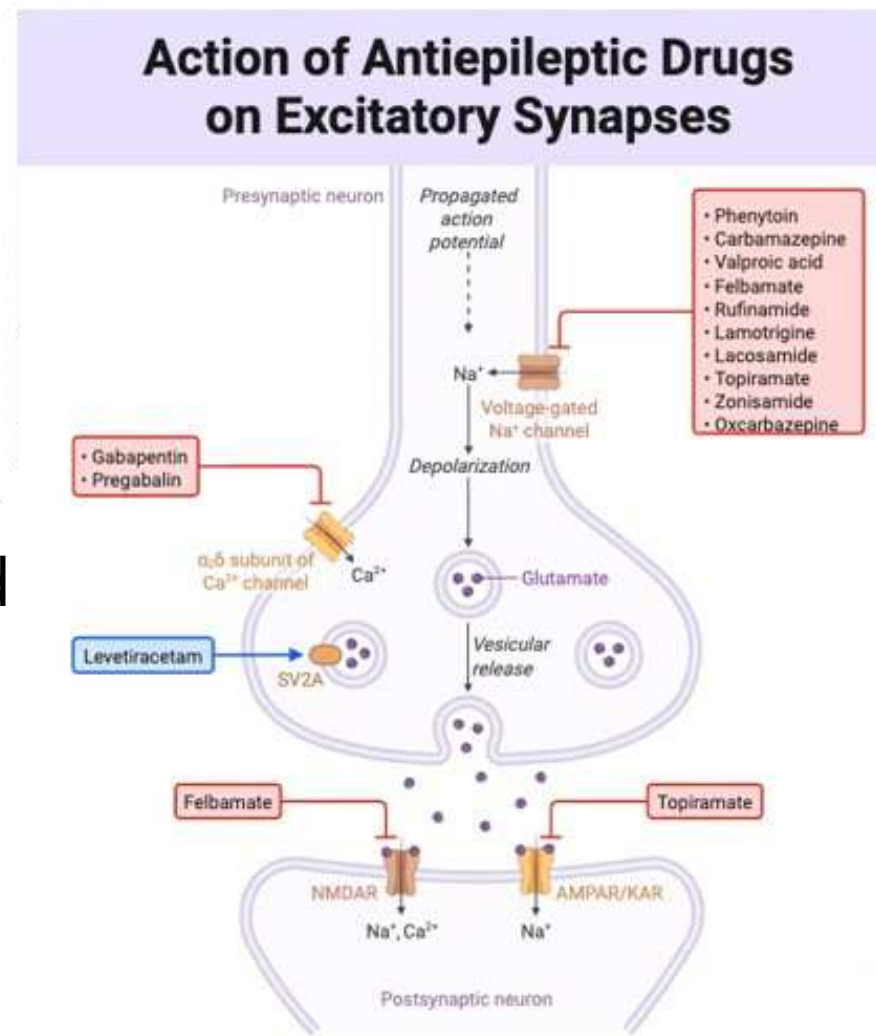
Epilepsy: a neurological disorder characterized by abnormal neuronal activity that can result in recurrent and uncontrolled seizures. Certain forms of epilepsy are associated with gain-of-function mutations that increase ion channel activity, leading to excessive neuronal excitability.



Commonly caused by **gain-of-function** mutations that increase ion channel activity, leading to excessive neuronal excitability.

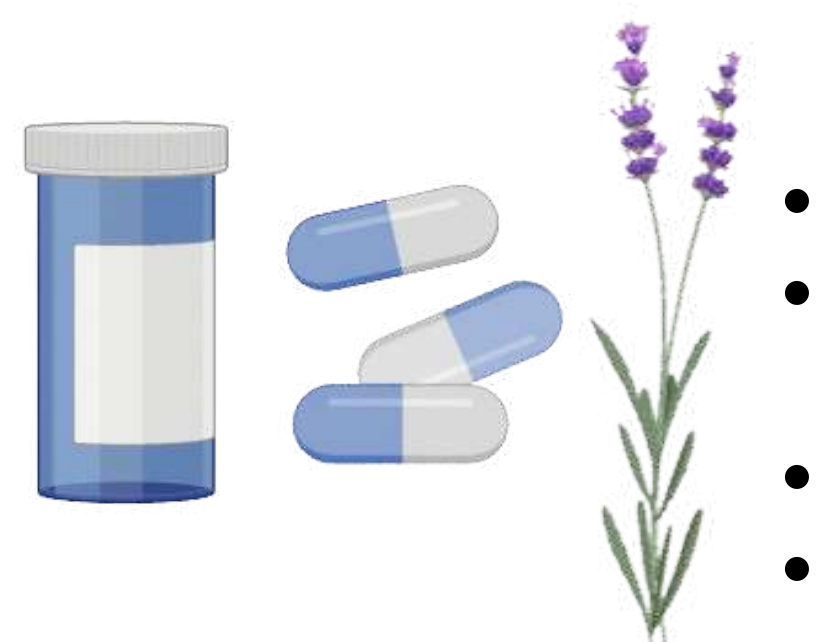
Current treatment: **Anti-epileptic drugs**

- High rate of drug resistance/ineffective in 30% of patients
- Slow down cognitive functions and chronic fatigue
- Behavioral toxicity and underlying depressive disorders
- Bone density loss and liver dysfunction



Proposed treatment: **Linalool**

- **Antioxidant** properties
- competitive antagonist of glutamate, limiting neural firing.
- combats oxidative stress
- low systemic toxicity



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SIGNIFICANCE

- Linalool will **reduce calcium levels** in neurons, rather than sodium levels, as calcium is strongly associated with hyperexcitability.
- Targeting intracellular calcium prevents neuronal damage.
- Potential therapeutic treatment for patients with seizures.

HYPOTHESIS

If:

Transfected SH-SY5Y cells are treated with linalool

Then:

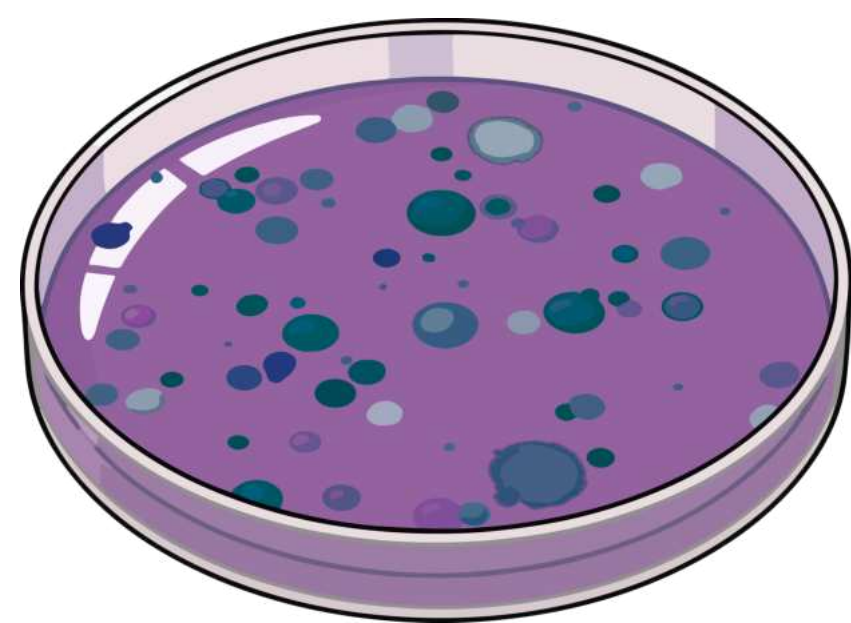
fluorescence dye will decrease, indicating reduced calcium influx.

Justification:

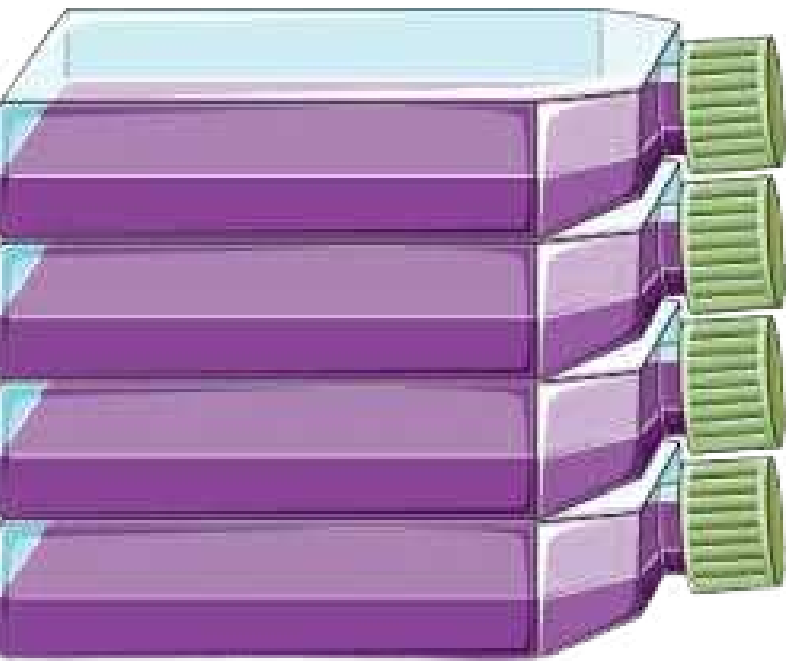
Linalool is known to have neuroinflammatory properties. Since Epilepsy is a gain-of-function mutation that increases neuronal excitability, linalool could reduce calcium levels.

METHODOLOGY

grow bacteria plates



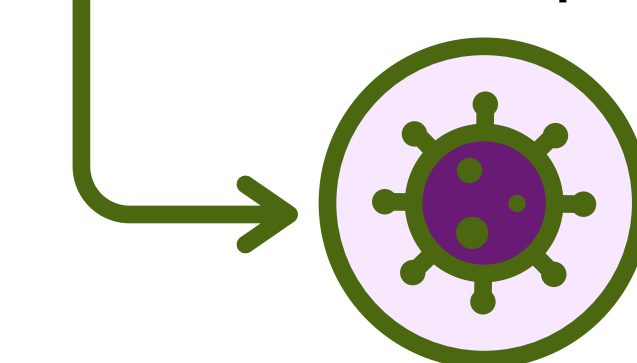
grow & culture cells



seed cells into well plates



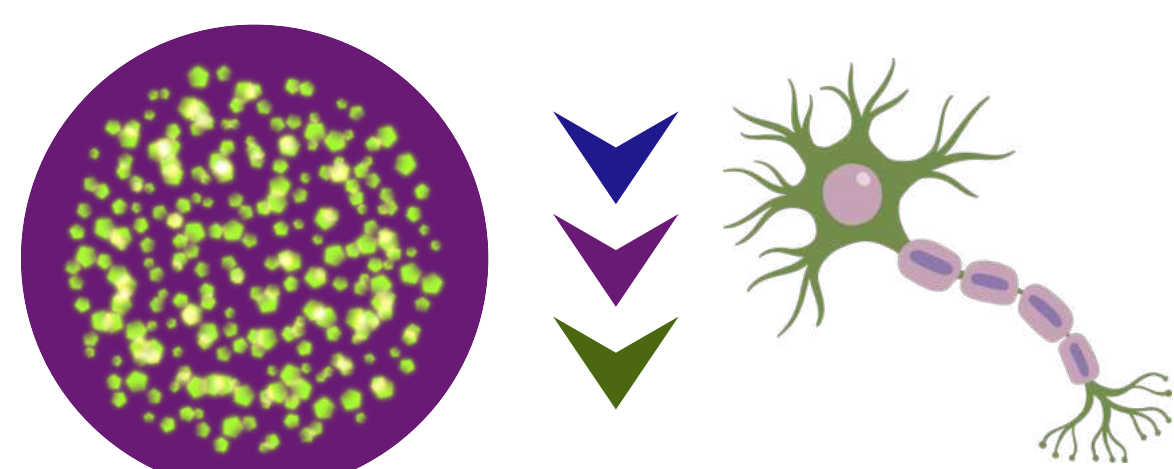
transfect with SCN2A plasmid



treat cells with linalool

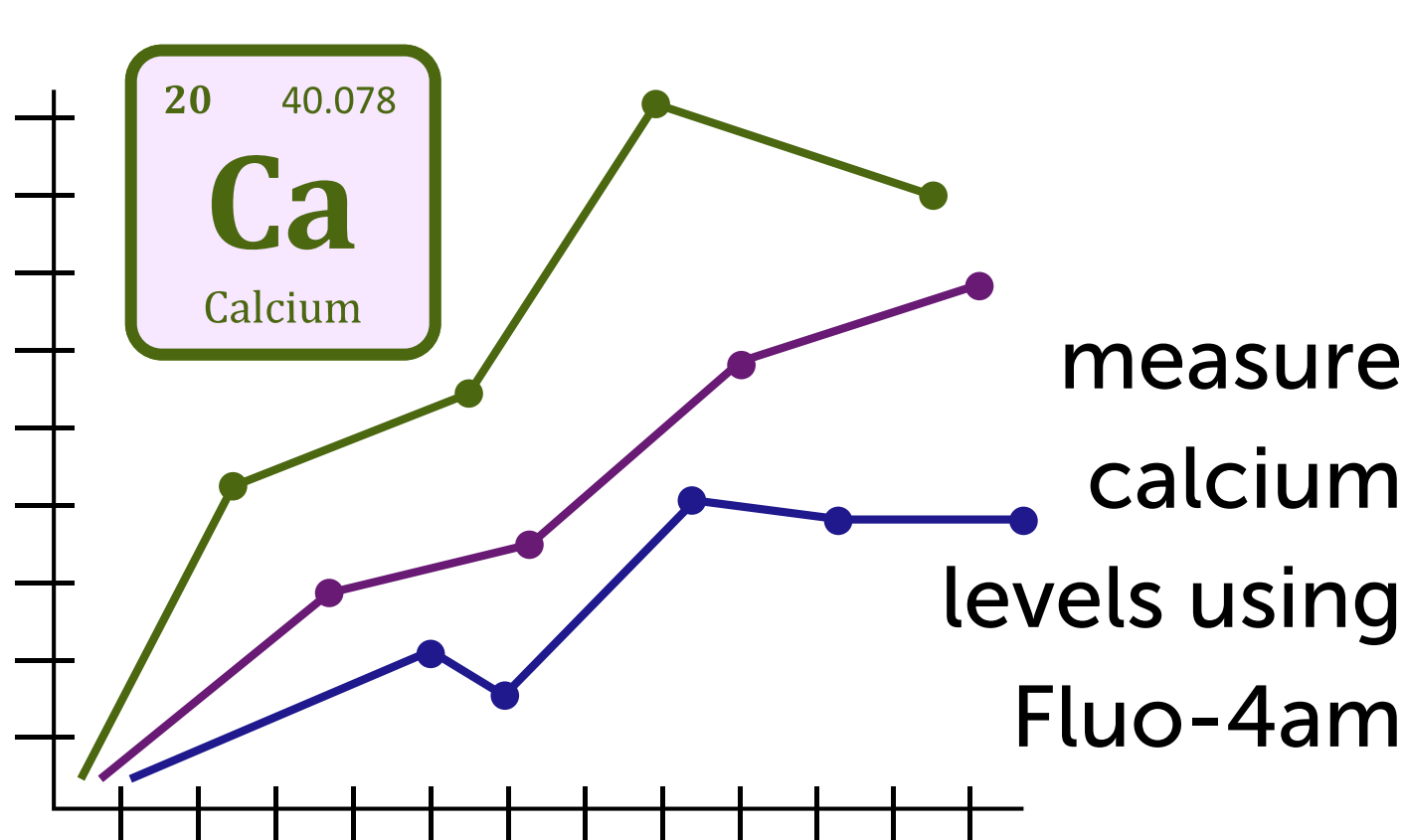


through stock solution



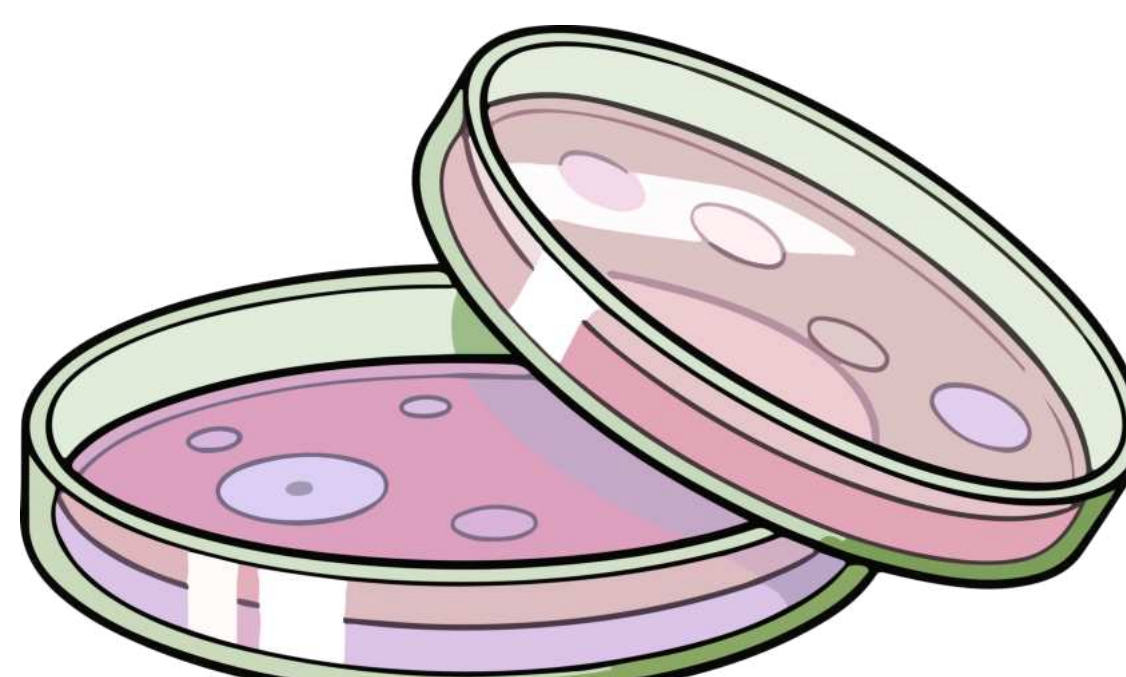
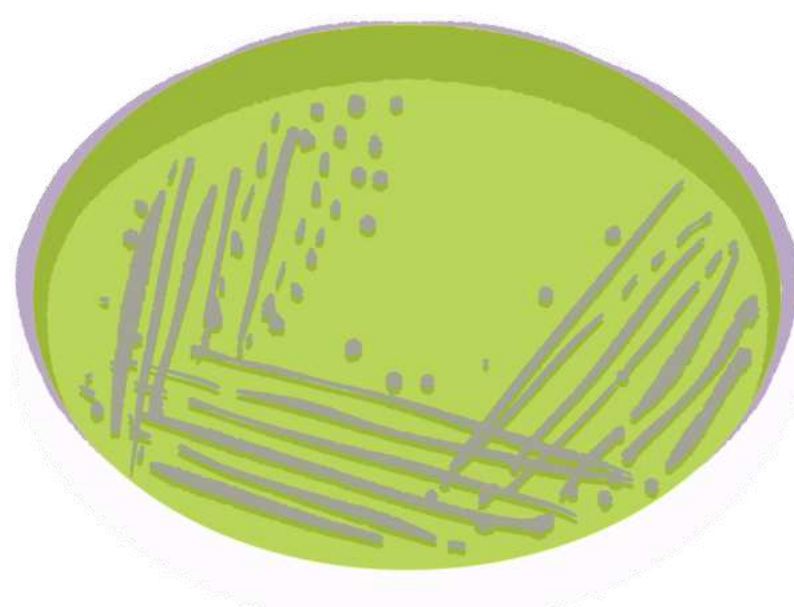
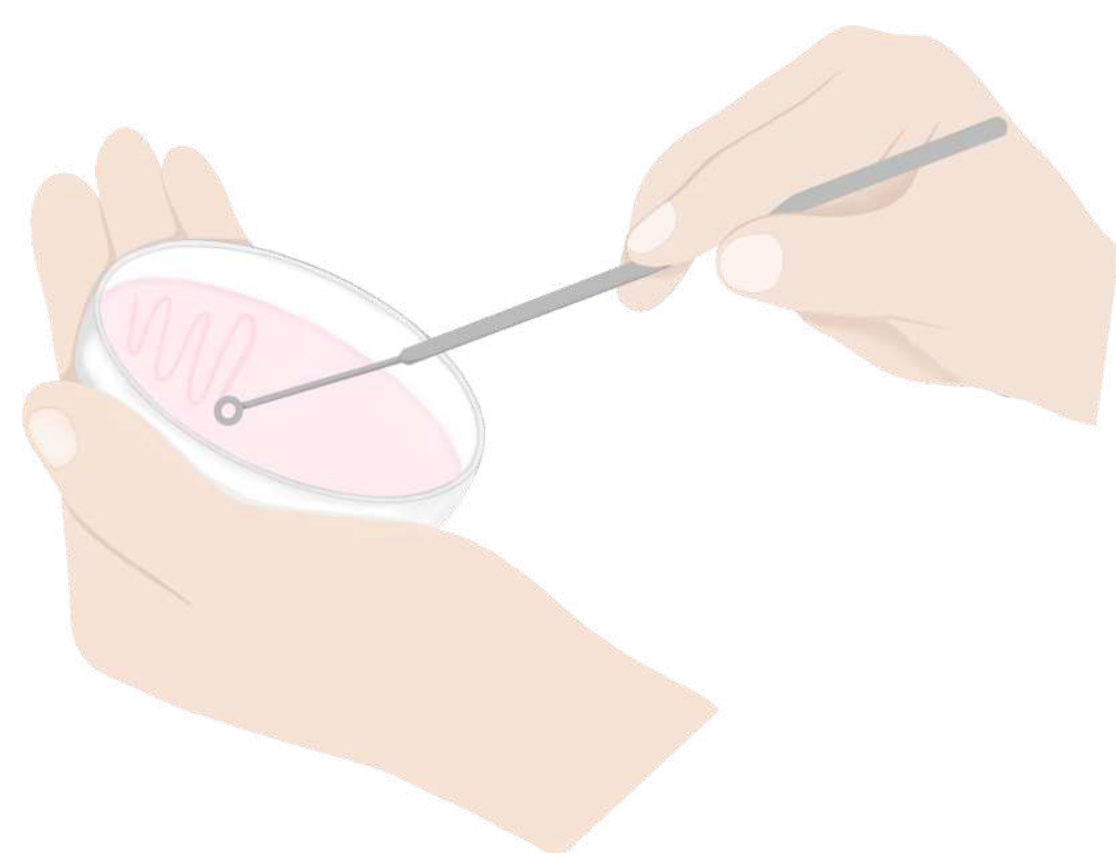
decrease in fluorescence

= decrease in Nav1.2



measure calcium levels using Fluo-4am

BACTERIAL CULTIVATION



Observed Outcome: Transformation of SCN2A-V4-mScarlet plasmid onto bacterial plates was successful.

The plates yielded distinct, isolated single colonies. Confirmed successful replication and provided the pure genetic material required for cell transfection.

RESULTS

DECREASE IN FLUORESCENCE INDICATES DECREASE IN CALCIUM

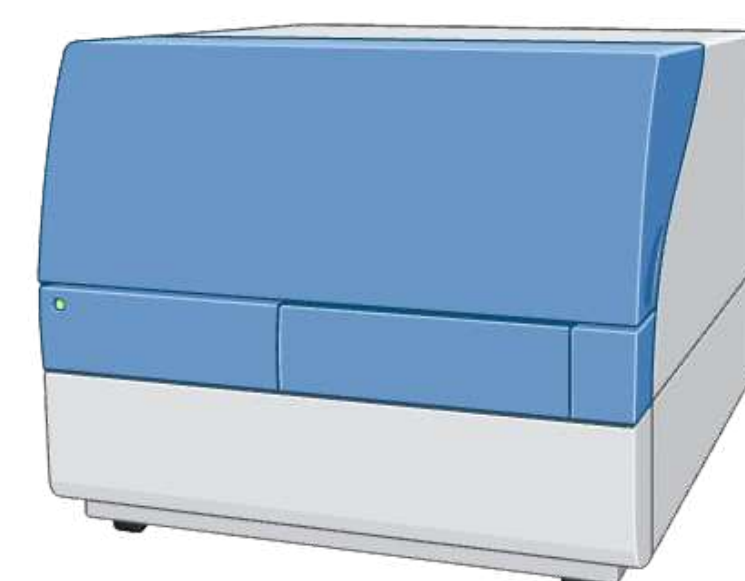
Data Normalization: All future raw data will be normalized against the un-transfected negative control to isolate the impact.

TEST: ANOVA will be applied to evaluate whether the reduction in calcium caused by linalool is statistically significant.

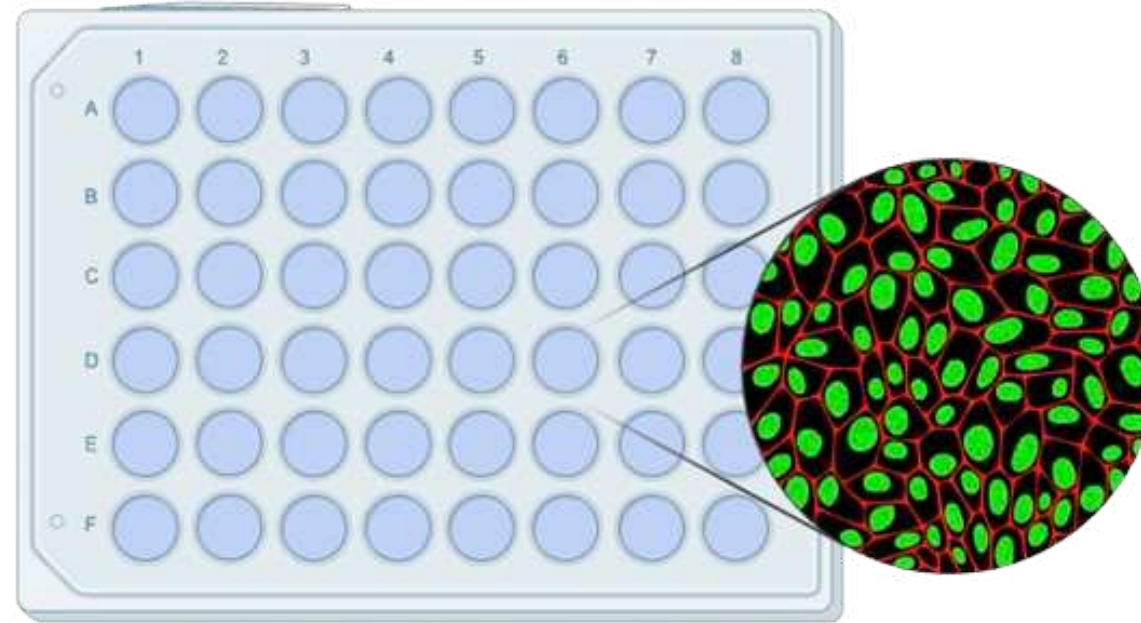
Negative Control Group: Expected to exhibit low baseline green fluorescence, normal homeostatic resting calcium levels in healthy SH-SY5Y cells.

Positive Control Group: Expected to exhibit a significant increase in fluorescence intensity

Experimental Group: Expected to exhibit a dose-dependent decrease in fluorescence intensity when exposed to varying working stocks of linalool.



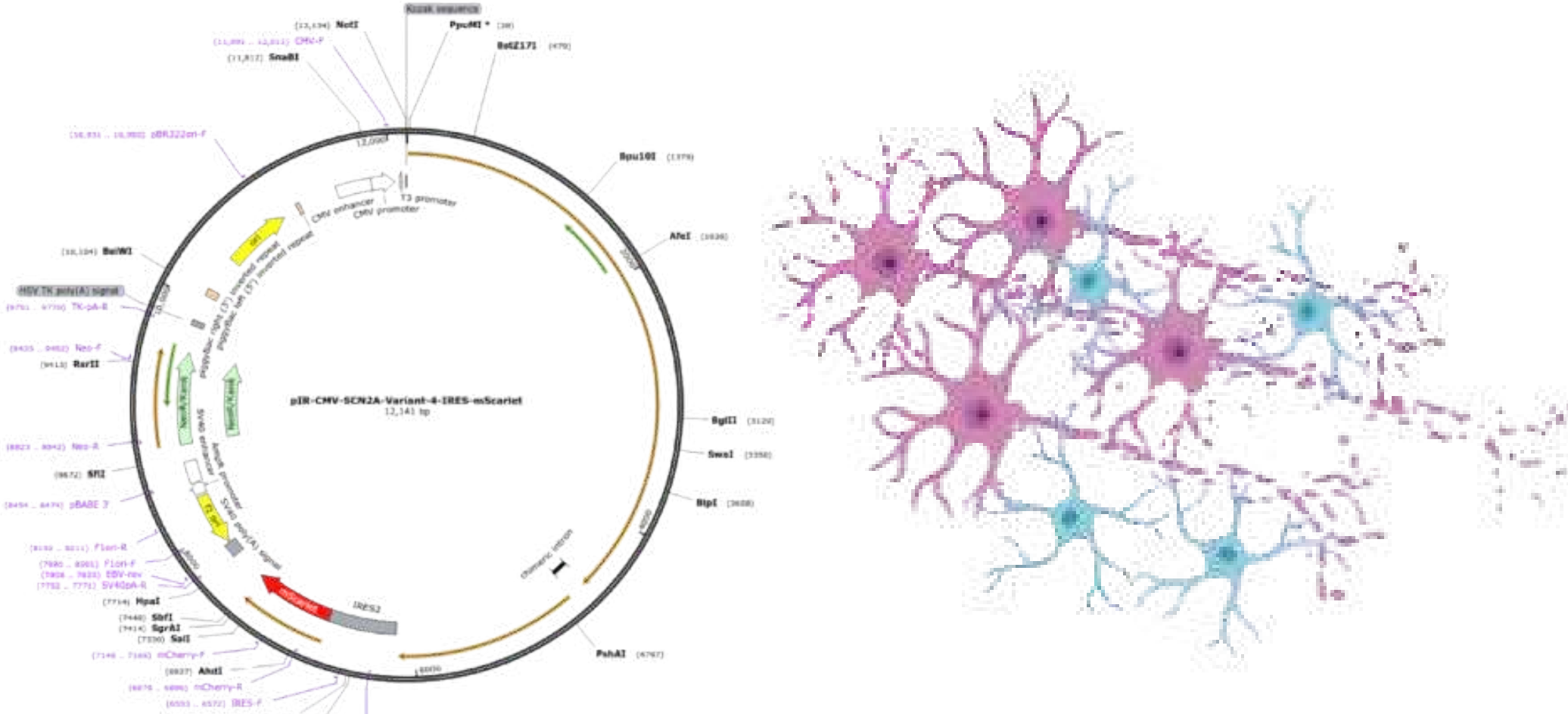
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DISCUSSION

INTERPRETATION

- **Validation of Mechanism:** If the experimental data match the expected model, a decrease in Fluo-4 AM fluorescence would directly confirm that linalool effectively blocks downstream calcium influx.
- **Targeted Action:** This reduction suggests that linalool modulates cellular hyperexcitability caused by the Nav1.2 gain-of-function mutation.
- **Neuroprotection Potential:** Because sustained intracellular calcium overload triggers apoptotic pathways, decreasing calcium influx implies that linalool provides direct neuroprotection to seizure stress.



NEXT STEPS/FUTURE WORK

Complete Current In Vitro Pipeline:

- **Recover Plasmid DNA:** Purify the replicated SCN2A-V4-mScarlet genetic material isolated from the single bacterial colonies.
- **Transfect & Treat:** Successfully transfect the seeded SH-SY5Y neuroblastoma cells and introduce varying working stock solutions of linalool.
- **Quantify Influx:** Execute the Fluo-4 AM fluorescent assay protocol under live-cell microplate reader to gather raw quantitative data.
- **Expand Dosimetry:** Test a broader range of linalool concentrations to determine the exact median effective dose and evaluate potential cellular toxicity thresholds.
- **In Vivo Modeling:** Eventually transition the treatment model into animal studies to test whether linalool can cross the blood-brain barrier and successfully reduce physical behavioral seizures.

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